L'ORÉAL BEAUTY TECH

L Pro

AI-Powered Hair Care Machine

Beauty should never depend on
physical ability, circumstance,
or access to assistance.

Executive Summary: An AI-Powered Hair-Care Machine Making Personalized Daily Care Accessible to Everyone

Market Overview

- **Promising Hair Care Market** with Strong Growth Potential
- **Older adults**, **pregnant women**, and **people with disabilities** still lack suitable hair-care solutions and often rely on hands-on assistance

Competitive Positioning

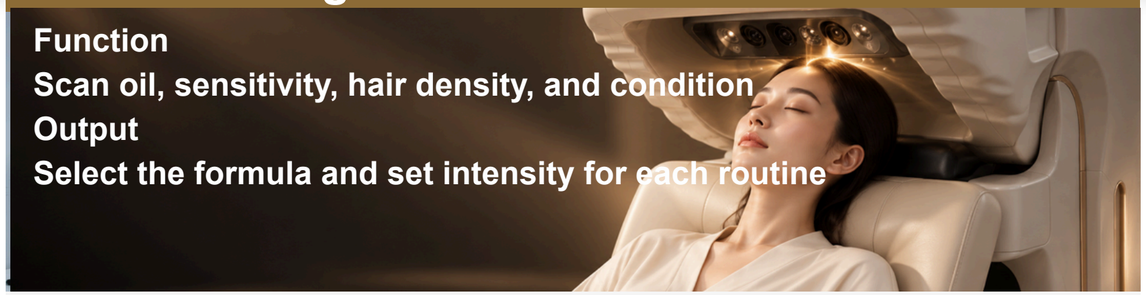
- **Panasonic** and **Emerging Hair-Washing Robots** Focus Primarily on **Automated Washing**
- **LPro Differentiates** Through an **Accessible, AI-Powered Full-Care System** Integrating **Diagnosis, Treatment, Washing, and Drying.**

Company Analysis

- **L'Oréal** Combines **Beauty Science**, **AI Diagnostics**, and **Global Consumer Expertise**
- **LPro** Extends These Strengths into **Accessible**, **Personalized**, and **Automated Hair Care**

L Pro brings professional diagnosis and treatment into one adaptive and personalized home routine

AI Diagnosis and Personalization



Function
Scan oil, sensitivity, hair density, and condition
Output
Select the formula and set intensity for each routine

Automated Care and Recovery

Function
Wash, massage, apply capsule care, use red light, and dry
Experience
Adjust intensity in real time and save household preferences

Connected Care Loop

Scalp scan → formula selection → automated wash and treatment → result feedback → updated user profile

Consumer Value
Convenient, consistent personalized care

+

Commercial Value
Device sales and recurring capsule demand

+

Strategic Value
Continuous insight into hair-care needs

Market Trends Analysis

Policy support, rising hair-care demand, evolving consumer needs, and proven beauty technology create a strong opportunity for LPro.

P- Political

8 Chinese ministries backing smart-home development

China's 2026 action plan encourages intelligent home appliances, personalized services, common technical standards, and broader consumer access.

E- Economic

Global hair care projected to grow 24% by 2030

The global hair-care market is forecast to reach £129 billion by 2030 as consumers spend more on scalp health and technology-enabled products.

S- Social

Scalp treatments are outpacing overall hair care

Scalp-treatment sales grew 4.2% in 2024, compared with 3.7% for the overall hair-care category. Consumers increasingly expect routines at home.

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T- Technological

AI scalp analysis has reached 9,000 salons, proving tehnological readiness

K-SPA and Water Saver also validate personalized treatment and efficient water-delivery technologies.

Sources: China Ministry of Commerce, Euromonitor via Vogue Business, WHO, L'Oréal K-SCAN, K-SPA and Water Saver.

Competitive Positioning

L Pro’s integrated care model creates a differentiated position with limited direct competition.

THREAT OF NEW ENTRANTS
LOW

Patents, water–electric safety requirements, long-hair handling, AI personalization, and after-sales capabilities create substantial barriers to replication.

BARGAINING POWER OF SUPPLIERS
MEDIUM-LOW

Buyers can compare LPro with salons and separate devices, but few alternatives provide the same level of automation, personalization, and accessibility.

THREAT OF SUBSTITUTES
LOW

Current alternatives address isolated needs, while LPro uniquely combines diagnosis, personalized treatment, washing, massage, and drying in one hands-free system.

BARGAINING POWER OF BUYERS
MEDIUM

Most hardware components can be sourced from multiple suppliers, while L’Oréal retains control over formulas, treatment capsules, and core beauty technology.

INDUSTRY RIVALRY
LOW-MEDIUM

Existing competitors focus on individual functions such as scalp analysis, washing, or drying, leaving limited direct competition in integrated full-cycle hair care.

Sources: Team product, technical architecture, IP landscape, and 4P strategy documents; L’Oréal K-SPA, K-SCAN

Competitor Analysis

LPro stands out by integrating diagnosis, personalized care, washing, and drying in one home-ready system.

	Competitor	Core Strengths	Key Gaps Relative to LPro
	Panasonic Head Care Robot	3D head scanning, automated shampooing, and robotic scalp massage.	Institutional prototype with a fixed reclining setup; limited home availability and no personalized daily-care ecosystem.
	QianLiang Hair Washing Robot	Hands-free washing and massage designed for commercial salon efficiency.	Focused mainly on washing; limited scalp assessment, formula personalization, drying, and household integration.
	JingXi AI Pulse Hair Washer	Automated pulse washing supports salons and head-spa operators.	Requires commercial installation and does not connect diagnosis, personalized treatment, and controlled drying.
	Zuowei ZW186Pro	Portable bedside cleaning supports elderly, disabled, and bedridden users.	Caregiver-operated and largely manual; lacks automated scalp care, treatment selection, and drying.
	Takara Belmont aqua pod	Automated salon shampooing uses rhythmic water flow to deliver consistent cleansing and reduce stylist workload.	Salon-only system focused on washing; lacks scalp diagnosis, personalized formulas, integrated treatment, drying, and home use.

Sources: Panasonic Technical Journal; Google Patents; Dreame official product pages, Takara Belmont Aqua product pages

Company Analysis

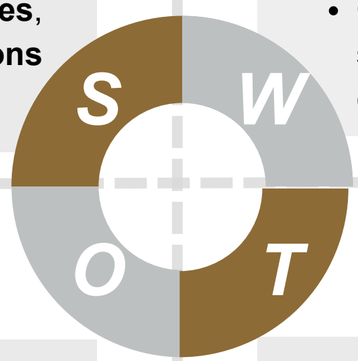
L’Oréal’s scale, hair-science expertise, and Beauty Tech provide a strong foundation for LPro, while hardware execution remains the key gap.

Strengths

- **€44.05B** in **2025 sales** and a **20.2% operating margin** provide the financial capacity to develop and scale a new device category.
- **Global omnichannel reach**, including salons, retail, pharmacies, and e-commerce, which now represents **more than 30% of sales**.
- **Strong hair science and Beauty Tech capabilities**, demonstrated by **K-SCAN** deployment across **9,000 salons** and continued investment in **AI diagnostics**.

Weaknesses

- **L’Oréal’s core capabilities** center on **beauty products, formulas**, and **professional services**, rather than **complex household appliances**.
- LPro requires **new capabilities in installation**, **maintenance**, **water–electric safety**, **repairs**, and **long-term hardware support**.
- Coordinating **hardware**, **capsules**, **customer data**, **salons**, and **after-sales service** increases **operational complexity**.



Opportunities

- **China** had **323.38 million people aged 60 or above** in 2025, expanding demand for **convenient and accessible personal-care solutions**.
- Around **1.3 billion** people **globally** experience **significant disability**, creating a **large underserved market** for more independent care.
- **L’Oréal** can use its **salon network** for trials and education, then **convert customers** through **e-commerce**.

Threats

- **Misuse of AI-generated customer data** could quickly **damage brand trust**. L’Oréal itself identifies **AI bias, transparency, privacy, and intellectual-property concerns** as **relevant risks**.
- Appliance and robotics companies may enter with **stronger manufacturing**, **engineering**, and **hardware-service capabilities**.

Sources: L’Oréal 2025 Annual Results and Annual Report; L’Oréal K-SPA and K-SCAN; National Bureau of Statistics of China; World Health Organization.

Strategic Overview

Transforming Hair Care from Assisted Service into an Independent, Personalized Daily Experience via Artificial Intelligence.

Mission

To make **professional**, **personalized hair care** accessible to **everyone**. **LPro** empowers people who face **physical**, **mobility**, or **situational barriers** to **care for themselves** with **greater independence**, **comfort**, and **dignity**—while giving every user a **smarter**, **easier**, and **more consistent** way to **experience beauty** in **everyday life**. **Beauty should never depend on physical ability, circumstance, or access to assistance.**

Vision

A future where hair care adapts to every person.

Remove physical, situational, and service barriers to professional hair care—making beauty more independent, inclusive, and effortless across homes and care environments.

ACCESSIBLE • PERSONALIZED • INDEPENDENT

Strategic Objectives

01 BUILD AN INCLUSIVE EXPERIENCE

Adapt to different body positions, mobility needs, hair lengths, and environments through flexible positioning and multimodal interaction.

02 DELIVER END-TO-END CARE

Integrate scalp observation, personalized washing, massage, treatment, and assisted drying into one seamless automated journey.

03 CREATE A SCALABLE ECOSYSTEM

Connect intelligent devices, personalized capsules, digital care profiles, and professional service networks to drive continuous use.

What Success Looks Like

GREATER INDEPENDENCE

Complete more of the routine with minimal assistance.

CONSISTENT CARE QUALITY

Reliable cleansing, comfort, personalization, and full-length hair coverage.

10 LAUNCH CITIES

Up to 100 experience and evaluation touchpoints.

SUSTAINABLE ENGAGEMENT

Repeat use supported by care profiles, capsules, and connected services.

1

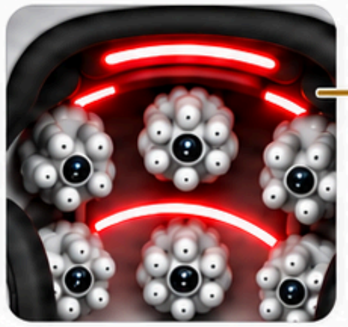
Product Strategy

L Pro combines intelligent scalp analysis, adaptive full-length care, and accessible multimodal control in one personalized system.

1 INTELLIGENT SCALP WASH & RED-LIGHT CARE

Camera-based scalp imaging, fluid-dispensing silicone contact heads, and a dedicated red-light care stage.

✓ **Observe, cleanse, and massage the scalp in one workflow.**



AI-ENABLED

AI analyzes scalp images and supports personalized care settings.

2 LONG-HAIR ZONAL CARE

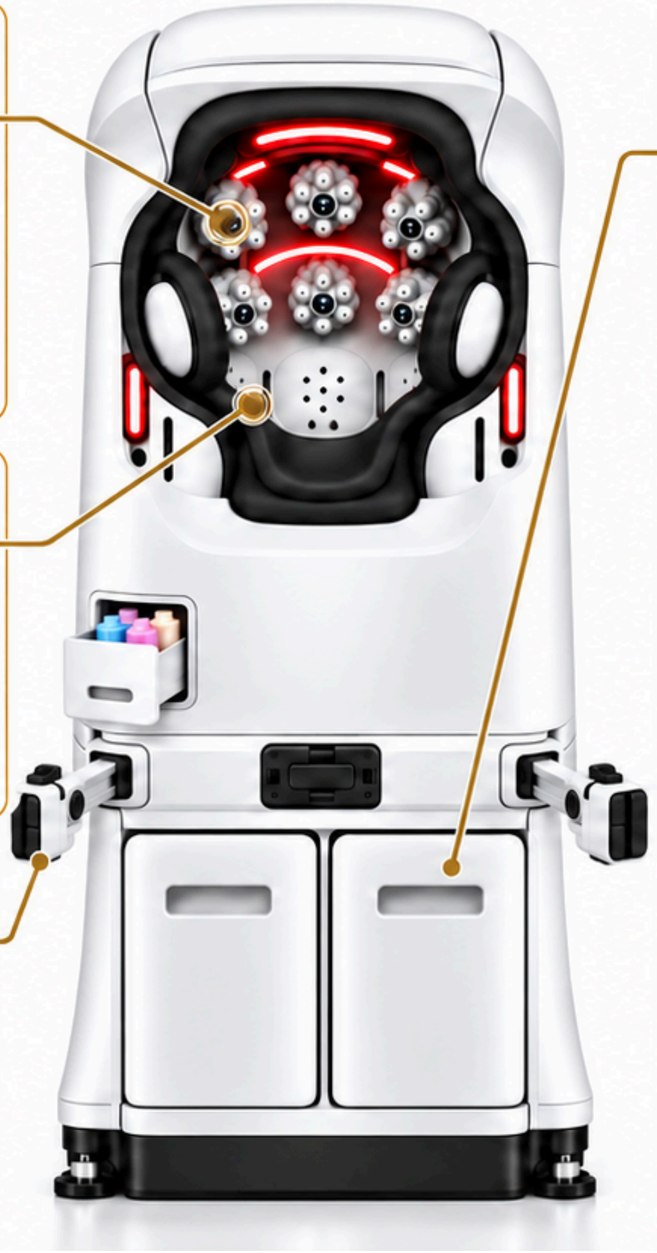
A vertical channel organizes long hair; alternating air cushions apply gentle pressure; outlets deliver treatment and air slots support drying.

✓ **Extends care from roots to ends while reducing manual hair handling.**

3 MULTI-SCENARIO ADAPTATION

Powered height, fore–aft, and internal–angle adjustments, plus wheelchair–handle connectors.

✓ **Adapts to chairs, wheelchairs, and bedside use.**



4 WATER & FORMULA MANAGEMENT

Separate clean– and waste-water tanks, pull-out refilling, and dedicated shampoo, conditioner, and treatment reservoirs.

✓ **Keeps the full care sequence continuous with less manual refilling.**

5 VOICE, GESTURE & TOUCHSCREEN CONTROL

Start or stop by voice, operate hands-free by gesture, or select programs and view status on the rear display.

✓ **Offers multiple control paths for different user abilities.**

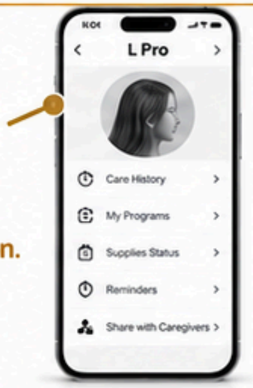
AI-ENABLED

AI supports voice and gesture recognition.

6 APP PERSONAL CARE PROFILE

Stores scalp images, care history, preferences, device status, supply levels, and reminders; authorized records can be shared.

✓ **Restores preferred settings and supports family or caregiver coordination.**



AI-ENABLED

AI uses care history to personalize settings and recommendations.

2

Position Strategy

L Pro is positioned as an inclusive, AI-powered hair care system that makes independent beauty care accessible to more people.

PREGNANT USERS

BEFORE

Awkward positioning and manual hair handling

WITH L PRO

Supported, leak-resistant automated care

OLDER ADULTS

BEFORE

Transfers and caregiver-intensive washing

WITH L PRO

Gentle care without leaving the wheelchair

USERS WITH LIMITED MOBILITY

BEFORE

Fixed controls limit independent use

WITH L PRO

Adaptive positioning and voice control

USERS WITH COGNITIVE SUPPORT NEEDS

BEFORE

Multi-step routines create confusion

WITH L PRO

Preset programs and clear guidance

Price Strategy

L Pro adopts a premium professional positioning, offering purchase, rental, and single-session options to meet different customer needs.

The prices below are proposed for market testing and do not represent official L'Oréal pricing.

OWNERSHIP
Purchase

HOME

¥29,800

Device • In-home fitting & installation
Personal contact set • 20 care sessions
2-year warranty

For long-term independent care at home

INSTITUTION

¥39,800

Multi-user turnover sets • Staff training
Quarterly maintenance in Year 1
2-year warranty

For senior care, maternity, hospitals & salons

LOWER UPFRONT COST
Rental

HOME

¥999 / month

3-month minimum • ¥499 installation
¥3,000 refundable deposit
Consumables sold separately

For recovery, temporary care or trial before purchase

INSTITUTION

¥1,999 / month

24-month contract
Installation • Training • Maintenance included
Consumables sold separately

Reduces initial procurement cost

TRY BEFORE COMMITTING
Single Experience

¥99 first visit

¥149 regular

Available through partner locations
Positioning • Automated wash & care
Basic finishing

Experience L Pro before purchasing or renting

CARE CONSUMABLES

Home: ¥199 / 20 sessions | Institution: ¥600 / 100 sessions

Approx. ¥9.95 vs. ¥6 per session; actual use depends on the selected program.

APP INCLUDED

Scalp records, care history, saved preferences, device control and maintenance reminders.
Voice, gesture and emergency stop controls included at no extra charge.

LAUNCH OFFER

Price maintained; added value through consumables and service.
Early home users: +20 sessions and one in-home maintenance visit.
Institutional clients: one complimentary staff refresher training.

Place Strategy

L Pro adopts a premium professional positioning, offering purchase, rental, and single-session options to meet different customer needs.

PHASE 1 | MARKET ENTRY

Build experience, institutional sales, installation and in-home service capabilities in the first five cities.

Shanghai	Launch events, training hub, salons, maternity centers and senior care
Beijing	Senior care and hospital partnerships; family experience locations
Guangzhou	Professional salons, maternity centers and consumables distribution
Shenzhen	Demonstrate voice control and App functions; convert trials into sales and rentals
Hangzhou	Convert Xiaohongshu, short-video and WeChat interest into appointments

PHASE 2 | REGIONAL EXPANSION

Enter after the trial, installation and maintenance processes have been validated in Phase 1.

Chengdu & Chongqing	Local experience and service hubs for Southwest households and institutions
Nanjing & Suzhou	Extend Yangtze River Delta coverage across maternity, senior care and home rental
Wuhan	Central China experience and training hub for institutional and regional service partners

RECOMMENDED SETUP PER CITY*

- **4** L'Oréal Professionnel partner salons
- **3** Maternity care centers
- **2** Senior care institutions
- **1** Hospital evaluation partner
- **1** Authorized installation & maintenance team

10-CITY RECRUITMENT TARGET

- **40** Salon experience locations
- **30** Maternity care centers
- **20** Senior care institutions
- **10** Hospital evaluation partners

Up to 100 experience, use or evaluation locations

Hospital evaluation partners are not consumer appointment locations.

ONE DIGITAL ENTRY, LOCAL DELIVERY

Website and WeChat Mini Program: city selection, nearby locations, appointments, purchase, rental and after-sales.

Address verification confirms service coverage before an order is accepted.

Installation and maintenance are initially limited to covered city zones; surrounding districts open gradually.

Promotion Strategy

L Pro combines trusted content with hands-on experience to turn awareness into trial.

1 SOCIAL MEDIA SEEDING

30 creators • 12 short videos

ACTIVATION

Hair, maternity and accessible-living creators share authentic L Pro experiences across Xiaohongshu, Douyin and WeChat Channels.

OBJECTIVE

Build credibility and drive trial bookings.

2 EXPERT DEMOS & LIVESTREAMS

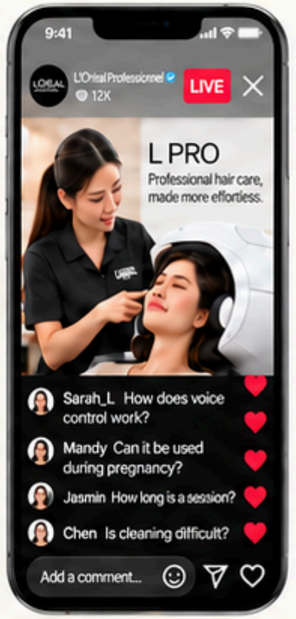
Biweekly live sessions

ACTIVATION

L'Oréal Professionnel stylists demonstrate operation, formula use and cleaning through live Q&A.

OBJECTIVE

Reduce uncertainty before trial or purchase.



3 10-CITY EXPERIENCE EVENTS

20 in-person events

ACTIVATION

Partner salons, maternity centers and senior care institutions host fit checks and hands-on trials.

OBJECTIVE

Convert experience into purchase or rental

4 LAUNCH & REFERRAL INCENTIVES

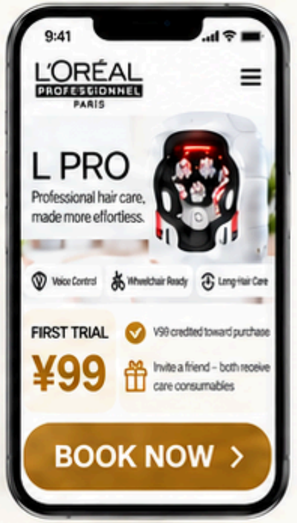
¥99 first trial

ACTIVATION

Credit the trial fee toward purchase and add consumables, maintenance or staff refresher benefits.

OBJECTIVE

Accelerate conversion and reward successful referrals.



Implementation Plan

L Pro will validate the product in 2026, launch in five cities in 2027, scale to ten cities in 2028, and prepare for profitable expansion by 2030.

WORKSTREAM	2026	2027	2028	2029	2030
PRODUCT & SAFETY Engineering • usability • compliance	Prototype engineering & safety validation		Design freeze & pilot manufacturing	Reliability upgrades & next-generation modules	
AI & DATA Scalp analysis • personalization • governance	Data governance & model baseline	AI-assisted scalp insights & care profiles		Personalized routines & predictive maintenance	
PILOT & EVIDENCE Users • caregivers • institutions	Controlled usability and caregiver pilots		5-city field validation	10-city outcomes and service-quality evidence	
MARKET & CHANNELS Experience • sales • rental	Partner recruitment & launch preparation	Phase 1: 5-city entry	Phase 2: 10-city coverage	Channel productivity & regional replication	
SERVICE & OPERATIONS Installation • training • maintenance	Service standards, training and parts readiness		Local installation teams & SLA tracking	Scaled maintenance, rental recovery & consumables	

2026	2027	2028	2029	2030
VALIDATE	**PILOT LAUNCH**	**SCALE**	**OPTIMIZE**	**REPLICATE**
3 pilot sites ≥100 test users ≥85% task completion Safety & privacy gate passed	5 cities ≥50 experience locations ≥1,000 trials Installation ≤7 days	10 cities Up to 100 locations ≥5,000 cumulative trials Device uptime ≥95%	NPS ≥50 Repeat usage ≥60% Consumables attach ≥70% Service response ≤24 hours	Profitable unit economics Maintenance cost ↓20% ≥90% service coverage Expansion playbook ready

Risk Mitigation, Implementation Plan & Expected Outcomes

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Balancing reliability, rollout, and growth for L PRO.

Risk & Mitigations

1. **Product Risk:** Washing, care, and drying may face reliability or hygiene issues.
2. **Adoption Risk:** High upfront prices and unfamiliar routines may slow uptake.
3. **Delivery Risk:** Manufacturing and after-sales overruns may compress margins.

Product Reliability & Safety

- Validate functions, cleaning, and safety controls.
- Refine reliability through pilots and user training.

Customer Adoption

- Use 20 demonstration units for hands-on trials.
- Offer purchase and rental options; validate both segments.

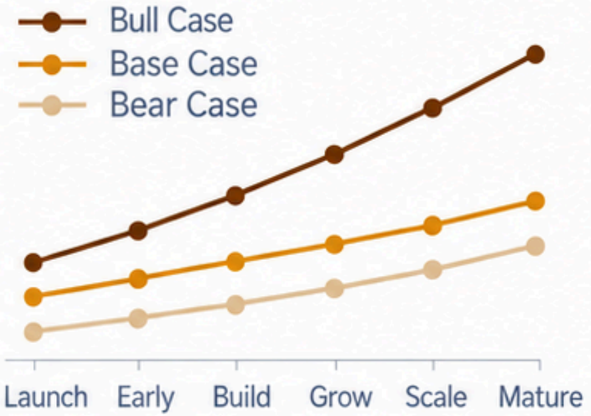
Cost & Delivery

- Confirm supplier quotes and alternative component sources.
- Phase production around orders, cash flow, and service capacity.

Implementation Plan

	Year 0	Year 1	Years 2–3	Years 4–5
Product Setup • Validate core functions • Qualify suppliers				
Pilot & Launch • Deploy demo units • Train service teams				
Scale & Optimize • Expand proven channels • Improve service and repurchase				

Expected Outcomes



Illustrative paths; not quantified forecasts.

L PRO Growth Scenarios

Illustrative scenario drivers

- **Bull case:** Faster adoption and stronger consumables repurchase support growth.
- **Target case:** Revenue rises from RMB 8.42m to RMB 16.00m; annual ROI increases from 3.33% to 9.01% over Years 1–5.
- **Bear case:** Slower conversion, lower usage, and higher service costs constrain growth.

Appendix 1: Revenue Breakdown + Cost Breakdown

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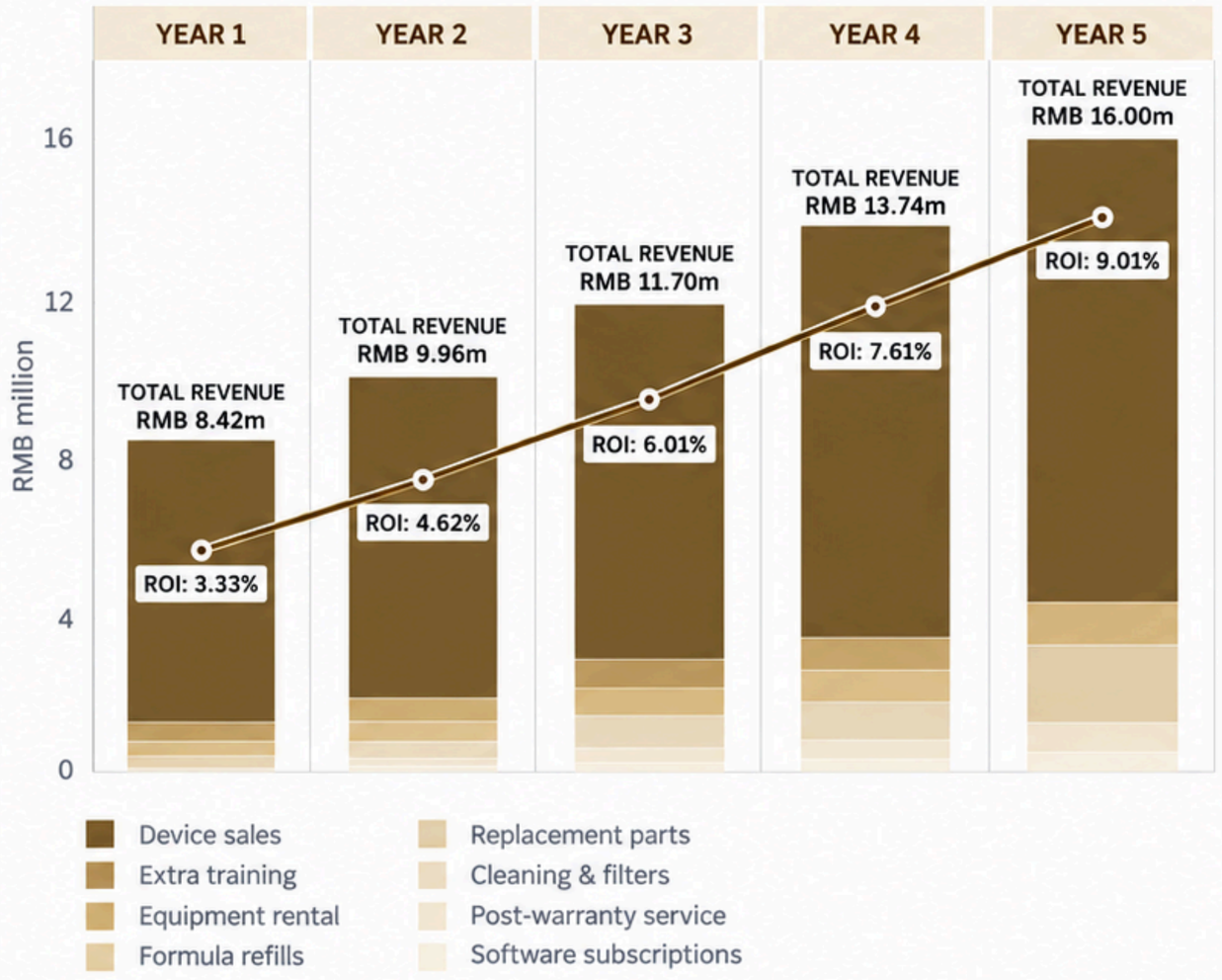
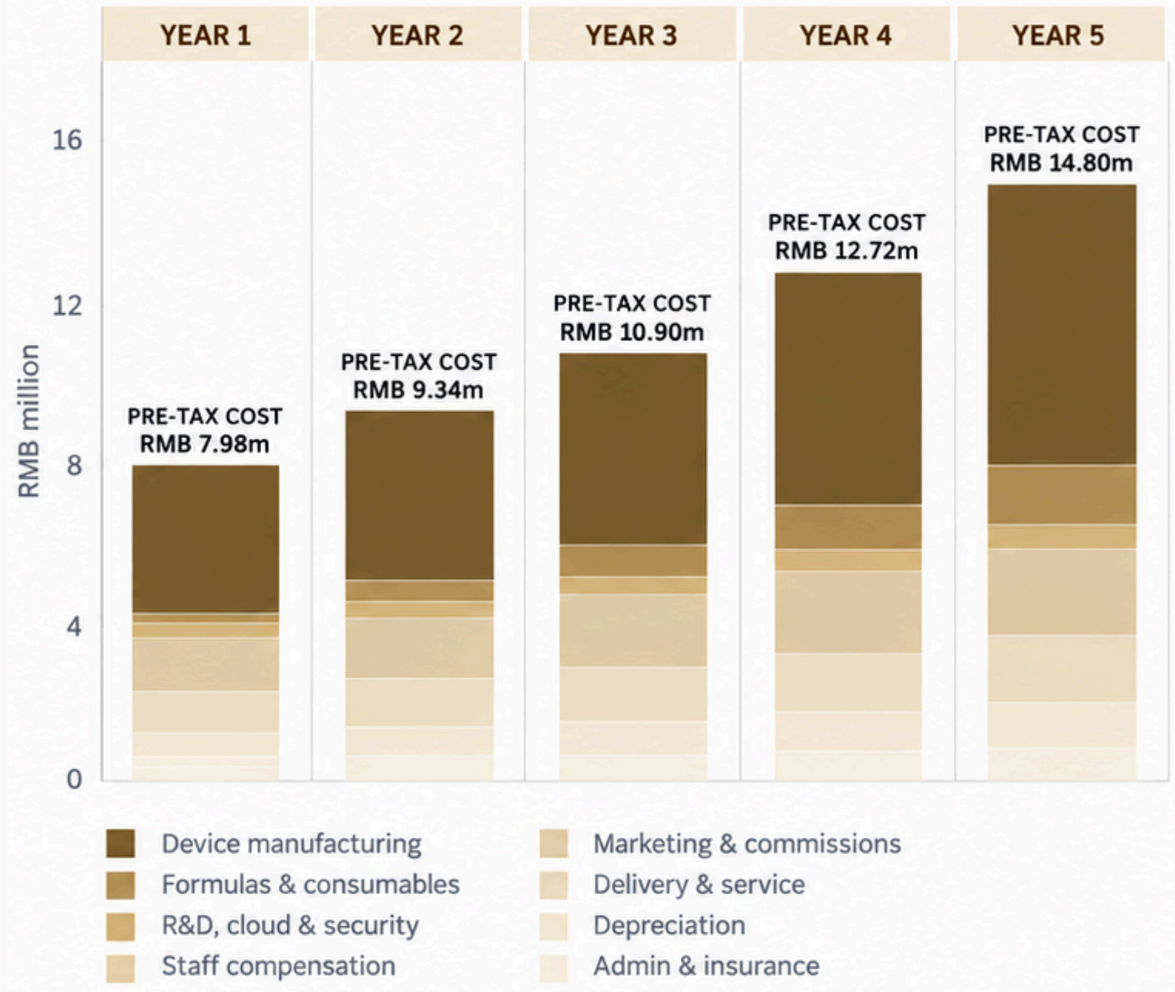
REVENUE BREAKDOWN (SALES & RECURRING INCOME)



COST BREAKDOWN (PRE-TAX EXPENSES)



Source: L PRO financial model | Target budget scenario | Revenue excludes VAT; costs exclude income tax.
ROI = annual net profit / RMB 10m initial investment. Cost categories are grouped; capital purchases enter expenses through depreciation. Brand and manufacturing costs require validation.